Geometry
Unit 2 Assessment

Name____________________
Date_________________
Period__________

Determine the type of reasoning for each statement.

	Statement	Type of Reasoning
1.	Sofia notices that the dogs on her street bark at a delivery truck, so Sofia concludes that all dogs bark at delivery trucks.	
2.	Florida requires all lawyers to pass the bar exam to practice. If Mark does not pass the bar, then he will not legally be able to represent as their lawyer.	
3.	The sum of the measures of two angles is $90°$, then the angles are complementary.	

4. An incomplete proof is shown. Complete the reason for step 2.

Statement	Reason
1. $m\angle TWX = m\angle YRG$	1. Given
2. $\angle TWX \cong \angle YRG$	2.

5. A series of steps for playing UNO is shown. The steps are not possible.

Steps
1. Green 7
2. Red 7
3. Yellow 9
4. Yellow 3
5. Wild Card
6. Red Reverse

a. Which step contains the error?

Error	Step________

b. What card, other than a wild card, would correctly complete the steps?

Card	

6. A statement is shown.

"If it is my birthday, then I will eat ice cream."

a. Write the converse of the statement.

Converse	

b. Write a counterexample that shows the converse is not true.

Counterexample	

7. Write each statement as an if-then statement.

Statement	If-Then
All Floridians have been to the beach.	
All four angles at the point of intersection of two lines are right angles.	

8. Complete the table for the conditional statement shown.

If Anthony has visited the Southernmost Point Buoy, then he has been to Key West.

Hypothesis	
Conclusion	

9. Write each conditional using “if and only if.”

If the sum of two angles is $180°$, then the angles are supplementary.

Biconditional	

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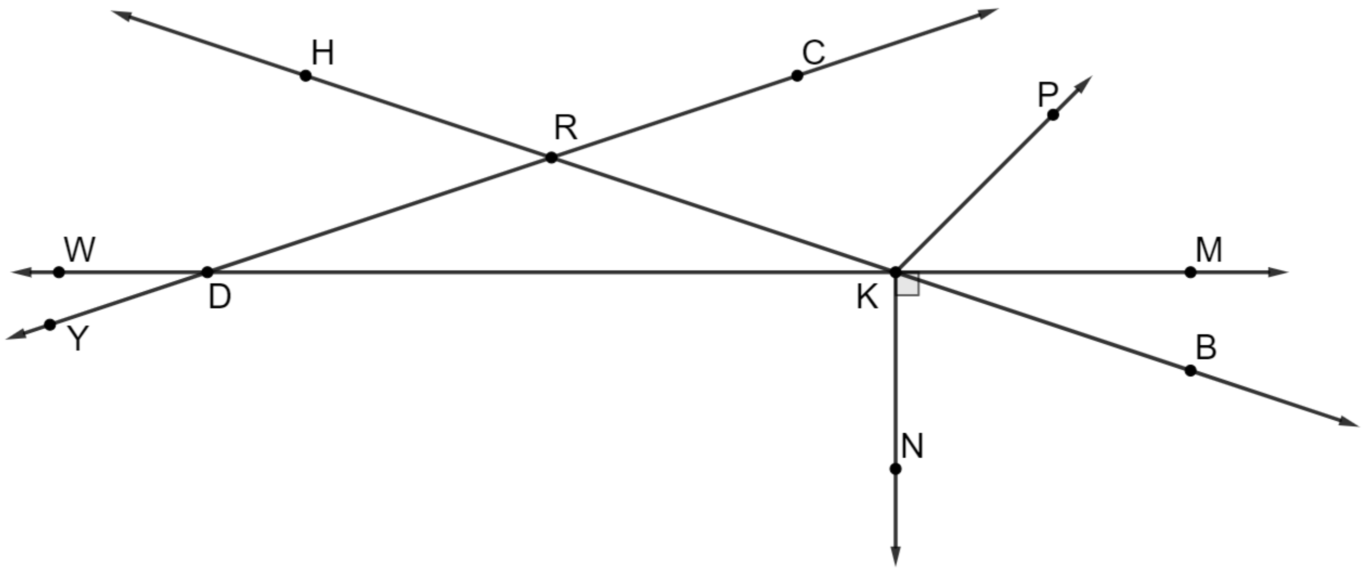
10. A diagram is shown.



a. Select all of the statements that are true.

- ☐ $\angle DRH \cong \angle KRC$
- ☐ $\angle HRC \cong \angle KRD$
- ☐ $\angle MKP \cong \angle BKN$
- ☐ $\angle NKB \cong \angle RKP$
- ☐ $\angle RDK \cong \angle RKD$
- ☐ $\angle WDY \cong \angle RDK$

b. An incomplete proof is shown. Complete the reasons for steps 2, 3, and 4.

Given: $m\angle MKB + m\angle BKN = 90°$

Prove: $m\angle RKD + m\angle BKN = 90°$

Statement	Reason
1. $m\angle MKB + m\angle BKN = 90°$	1. Given
2. $\angle MKB \cong \angle RKD$	2.
3. $m\angle MKB = m\angle RKD$	3.
4. $m\angle RKD + m\angle BKN = 90°$	4.

c. Complete the table for the conditional statement.

If $m\angle MKN = 90°$, then $m\angle MKB + m\angle BKN = 90°$.

Relationship to Conditional	**Statement**
Converse	
Inverse	
Contrapositive	

d. Explain why the conditional statement is biconditional.

11. A series of steps is possible in the game of UNO. Each statement represents a card played.

Statement	**Reason**
1. Blue 5	1. Given
2. Blue 2	2. Matches color
3. Yellow 2	3. Matches number

Without using a wild card, add two additional statements and provide the reason for the series of steps.

Statement	**Reason**
4.	4.
5.	5.

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